

CYBER INCIDENT INVESTIGATION USING NETWORK TRAFFIC CAPTURE

Hashan Kodippilige
Network Security -Project 3



INCIDENT OVERVIEW

- ACME Inc. reported suspected malicious activity
- CEO Sal Patel requested a forensic investigation to reconstruct the attacker's actions
- Victim system IP address -
 - 192.168.10.4
- Evidence provided –
 - Packet capture files (file1–file5.pcap)
 - FTP server log file (client_logs.log)

INVESTIGATION OBJECTIVES

- Analyze FTP server logs for suspicious activity
- Examine packet captures using Wireshark
- Identify attacker behavior and tactics
- Determine evidence of –
 - Reconnaissance, exploitation, lateral movement, and possible data exfiltration
- Map attacker activity to the Cyber Kill Chain, and MITRE ATT&CK framework
- Reconstruct a timeline of the attack



TOOLS USED

- **Linux/Kali Tools** - Log analysis and command-line investigation
- **Wireshark** - Packet capture and network traffic analysis

PART 1 - LOG ANALYSIS OVERVIEW

- Investigation of client_logs.log from ACME Inc.
- Goals -
 - Analyze log file using grep, awk commands
 - Identify suspicious FTP activity
 - Determine evidence of unauthorized access
 - Build timeline of events

PART 1 – LOG ANALYSIS DEMO

PART 2 - WIRESHARK ANALYSIS OVERVIEW

- Analyzed file1–file5.pcap
- Used Wireshark filters and protocol analysis
- Goals –
 - Identify attacker behavior
 - Detect scanning
 - Detect exploitation
 - Identify lateral movement and exfiltration

PART 2 - WIRESHARK ANALYSIS DEMO

ATTACK ANALYSIS SUMMARY

Evidence Source	Activity Identified	Cyber Kill Chain Stage	MITRE ATT&CK	Adversary Goal
Log Files	Multiple client connection attempts from 192.168.10.5	Reconnaissance / Initial Access	T1078 - Valid Accounts	Establish access to target system
Log Files	Successful anonymous FTP logins detected	Initial Access	T1078 - Valid Accounts	Gain unauthorized FTP access
Log Files	Repeated authentication activity without upload/download evidence	Persistence / Enumeration	T1021 - Remote Services	Maintain interaction with service
file5.pcap	ICMP ping sweep / host discovery	Reconnaissance	T1018 - Remote System discovery	Identify active hosts
file2.pcap	TCP SYN port scanning	Reconnaissance	T1046 - Network Service Scanning	Identify exposed services
file3.pcap	FTP interrogation and authentication	Initial Access	T1078 - Valid Accounts	Authenticate to FTP service
file4.pcap	HTTP enumeration and web probing	Reconnaissance	T1595 - Active Scanning	Discover hidden web resources
file1.pcap	Persistent TCP communication	Post-Access Activity	T1071 - Application Layer Protocol	Maintain communication with victim

ATTACK TIMELINE

Time 2026-05-04	Activity	Evidence Source	Attack Stage
16:34	Initial client connection from 192.168.10.5 detected	Log Files	Reconnaissance
16:37	Successful anonymous FTP logins observed	Log Files	Initial Access
17:49 – 17:55	Repeated FTP connections and authentication attempts	Log Files	Persistence / Enumeration
20:41	ICMP ping sweep detected between attacker and victim	file5.pcap	Reconnaissance
20:59	HTTP GET requests and web enumeration activity observed	file4.pcap	Reconnaissance
21:48	SYN-based port scanning against multiple services detected	file2.pcap	Reconnaissance
21:55	FTP interrogation and authentication traffic captured	file3.pcap	Initial Access
21:59	Persistent TCP communication established	file1.pcap	Post-Access Activity

KEY FINDINGS FROM LOG AND PCAP ANALYSIS

- Suspicious activity originated from - 192.168.10.5
- Evidence of —
 - Host discovery
 - port scanning
 - FTP enumeration
 - Web enumeration
 - successful anonymous FTP authentication
- Vulnerable FTP service identified —
 - vsFTPd 2.3.4
- Possible post-access activity observed
- No definitive evidence of data exfiltration identified

CONCLUSION

- Suspicious activity originated from internal host 192.168.10.5
- Evidence of host discovery, port scanning, FTP interrogation, and web enumeration was identified
- Multiple services including FTP, HTTP, HTTPS, SMB, and MySQL-related ports were targeted during scanning activity
- The attacker identified vsFTPd 2.3.4, a historically vulnerable FTP server version
- Anonymous FTP authentication was successfully observed
- Persistent post-access communication was observed
- No definitive evidence of successful data exfiltration was identified

RECOMMENDATIONS

- Disable anonymous FTP authentication
- Patch or remove vulnerable services such as vsFTPd 2.3.4
- Enforce stronger authentication policies
- Monitor internal network traffic
- Deploy IDS/IPS solutions
- Segment internal networks
- Restrict unnecessary services and open ports

REFERENCES

- TCM Security. (n.d.). *Invaluable log analysis tools: Sed, Awk, Grep, and RegEx*. TCM Security. <https://tcm-sec.com/log-analysis-sed-awk-grep-regex/>
- HackTricks. (n.d.). *Linux forensics*. HackTricks Wiki. <https://book.hacktricks.wiki/en/generic-methodologies-and-resources/basic-forensic-methodology/linux-forensics/index.html>
- Moore, K. (2023, November 19). *Analyzing PCAP files using Wireshark*. Medium. <https://medium.com/@viewshola/analyzing-pcap-files-using-wireshark-73fc1bef3c05>
- Wireshark Foundation. (n.d.). *Working with captured packets*. Wireshark User's Guide. https://www.wireshark.org/docs/wsug_html_chunked/ChapterWork.html
- Kinzie, K. (2022, August 19). *How to use Wireshark: Comprehensive tutorial + tips*. Varonis. <https://www.varonis.com/blog/how-to-use-wireshark>

The background is a dark blue gradient. In the corners, there are white line-art illustrations of circuit boards or neural networks, with lines and small circles representing nodes.

THANK YOU!